

# Gurram Naga Kumaraswamy

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## SUMMARY

Final-year Computer Science and Engineering undergraduate. Two completed internships shipping production platforms end to end across React, TypeScript, Python, FastAPI and PostgreSQL, plus measured LLM/GenAI security work in agent containment, concurrency control and application security.

## EDUCATION

**Indian Institute of Information Technology, Sri City**  
*Bachelor of Technology in Computer Science and Engineering*

August 2023 – May 2027  
Sri City, Andhra Pradesh

## EXPERIENCE

### Full Stack Developer Intern

May 2026 – August 2026

*Krea University*

*Remote*

- Shipped the pilot release of PRISM, a health-research data management platform, end to end across React, FastAPI and PostgreSQL, taking a live data-collection workflow from schema design to production UI.
- Secured sensitive research records by designing role-based access control (RBAC) and multi-factor verification aligned with India's Digital Personal Data Protection (DPDP) Act, 2023.

### Full Stack Developer Intern

January 2026 – April 2026

*Cymax*

*Remote*

- Built Unity Stream, a secure VR streaming platform, delivering in-browser immersive sessions with React and A-Frame/WebXR across multiple headset targets.
- Protected stream and session data with AES-256 encryption and built multi-headset session management for concurrent devices.

## PROJECTS

### Perimeter — LLM Agent Security Proxy

[github.com/kumarswamyg2005/Perimeter](https://github.com/kumarswamyg2005/Perimeter)

*Python, Model Context Protocol (MCP), DeBERTa, ONNX Runtime, Docker, pytest*

- Cut indirect prompt-injection success from 100% to 0% across a self-authored 213-payload, 9-technique, 1,264-trial benchmark by building a transparent MCP proxy that enforces capability scoping and taint-based containment on every LLM agent tool call.
- Proved through a 4-arm ablation that policy-level containment blocked 100% of attacks at 1/80th the latency of a DeBERTa ONNX classifier, holding 99% benign completion at +0.34 ms p95; shipped with 71 tests, mypy strict and CI.

### DesignDen — Custom Clothing E-Commerce Platform

[github.com/kumarswamyg2005/Designden](https://github.com/kumarswamyg2005/Designden)

*React, Redux Toolkit, Node.js, Express, MongoDB, Redis, Three.js, Docker*

- Built a full-stack MERN platform with 5 role-based dashboards (customer, designer, manager, delivery, admin) and a Three.js 3D studio for garment customization.
- Cut product-query latency by 97% with Redis caching and 14 MongoDB indexes, and shipped a Docker and GitHub Actions CI/CD pipeline gated by 84 passing Jest unit tests.

### SeatLock — Race-Condition-Safe Ticketing Platform

[github.com/kumarswamyg2005/SeatLock](https://github.com/kumarswamyg2005/SeatLock)

*React, TypeScript, Node.js, Prisma, PostgreSQL, Redis, Stripe*

- Guaranteed zero double-booking under concurrent load by combining atomic Redis TTL locks with PostgreSQL SELECT...FOR UPDATE transactions behind an interactive SVG stadium seat map.
- Demonstrated sub-5 ms race resolution (HTTP 200 vs. 409 Conflict) in an in-app concurrency test lab, and integrated Stripe checkout with JWT authentication.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, HTML/CSS

**Frontend:** React, Next.js, Redux Toolkit, Three.js, A-Frame/WebXR, Tailwind CSS, Vite

**Backend:** Node.js, Express, FastAPI, Socket.io, REST APIs, JWT Authentication

**Databases:** PostgreSQL, PostGIS, MongoDB, Redis, Prisma ORM

**GenAI and LLM Systems:** Model Context Protocol (MCP), prompt-injection defense, retrieval-augmented generation (RAG), vector search, ONNX Runtime, LLM evaluation

**Tools and Platforms:** Docker, Docker Compose, GitHub Actions (CI/CD), Git, pytest, Jest, Stripe, Vercel

**Concepts:** Concurrency Control, System Design, Application Security, AES-256 Encryption, Unit Testing